

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Electronics and Communication Engineering**  
**Academic Year: 2019 - 2020 (Even Semester)**

**Innovative Practices Description**

**UNIT I Feedback Amplifiers and Stability**

**Degree, Semester & Branch: IV Semester B.E ECE A**

**Course Code & Title: EC8452 Electronic Circuits II**

**Name of the Faculty member: Mrs.R.Chitra**

**Name of the topic: Effect of feedback on input impedance**

**Name of the Activity: Sketch Noting**

**Date & Duration: 04.01.2020 & 15 minutes**

**Description:**

Sketch Noting, in its purest form, is creating a personal visual story as one is listening to a speaker or reading a text. Sketch-notes can be any size, so invite students to cover whiteboards with everything they've learned in a unit or topic as a form of pre-assessment review.

**Goals (Learning Outcomes):**

1. The students will be able to learn the impact of negative feedback on input resistance.

**Use of appropriate methods:**

**Justification for choosing the topic using Sketch Noting activity:**

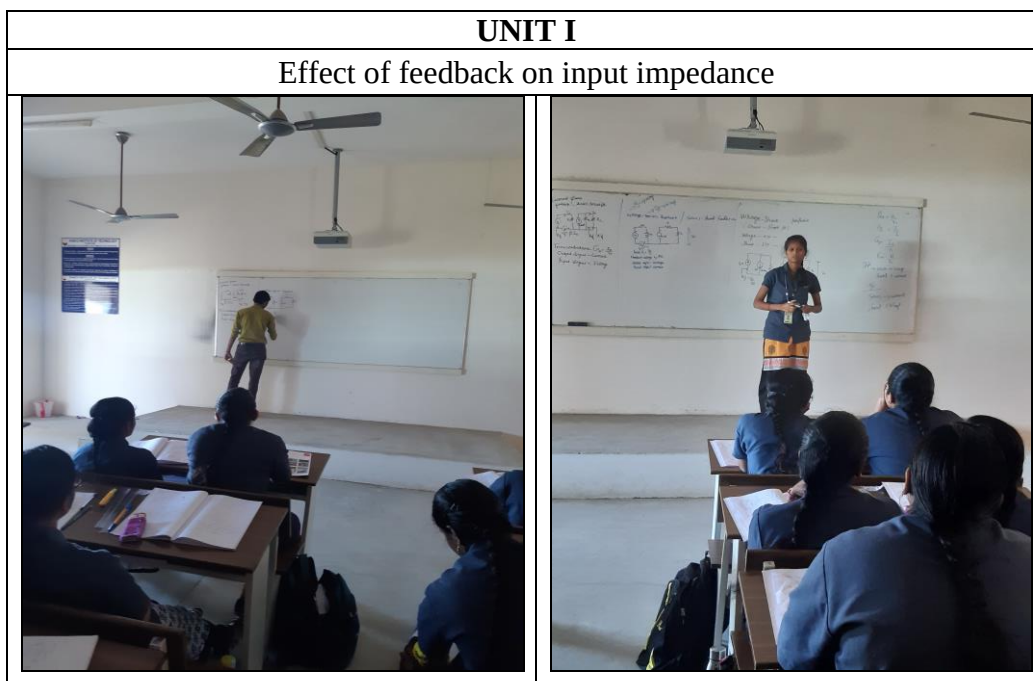
In unit I, there are four feedback topologies and need to analyze the impact of feedback on gain stability, bandwidth and input and output resistance using equivalent circuit of concerned feedback topology. As it may be confused, making it as a practice to draw the equivalent circuit is a healthy aspect to remind circuit diagram. As an exam point of view, it is frequently asked topic in University exam.

**Effective presentation**

**Implementation (Plan & Execution) with Proof:**

- ✓ Basically, sketch-noting makes the students to remember information stick to the brain.
- ✓ I have given five minutes to students to recall the concept involved in the given topic.
- ✓ After that, I invited 4 students by calling their register number randomly and asked each student to cover whiteboards with everything they've learned in a given feedback topology.

## Sample Images:



## Significance of Results:

### Assessment of Effectiveness/Success of the Activity:

- By conducting this activity, the students were able to recall the topic and they were able to draw the equivalent circuit of each feedback topology for given topic. Also, they can able to compare the differences between different feedback amplifiers.
- After conducting this activity, 60% of students were able to draw equivalent circuit for the next topic on Effect of feedback on output impedance while i am teaching during the class hour.

### Reflective Critique:

#### • Benefits:

- In this subject we need to draw the circuit using the symbol of electronic components. But, students failed to draw the symbols properly. Then I explained how to draw the circuit properly. Later they were able to realize their mistakes and it can be corrected.

#### • Challenges:

- Initially, the invited student felt shy to come to stage and draw the circuit. Later I motivated them to participate in this activity.
- I have planned to conduct this activity for 10 minutes. But, it took 15 minutes to complete this activity

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1            | 3   | 1   | 3   | 1   | 2   | -   | -   | 1   | -   | 2    | 1    | 1    |

*CO1: The students will be able to analyze the concepts of Feedback Amplifiers in various applications.*

### References:

1. <https://www.weareteachers.com/use-sketchnotes-in-the-classroom>
2. Sedra and Smith, "Micro Electronic Circuits"; Sixth Edition, Oxford University Press, 2011.
3. S.Salivahanan, N Suresh Kumar, "Electronic Circuits II" Mc Graw Hill Education.

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**UNIT I Feedback Amplifiers and Stability**

**Degree, Semester & Branch: IV Semester B.E ECE A**

**Course Code & Title: EC8452 Electronic Circuits II**

**Name of the Faculty member: Mrs.R.Chitra**

**Name of the topic: Analysis of series-series Feedback Amplifiers**

**Name of the Activity: One Minute Paper**

**Date & Duration: 07.01.2020 & 5 minutes**

**Description:**

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, Students must first organize their thinking to rank the major points and then decide upon a significant question. It is thus a very adaptable tool.

**Goals (Learning Outcomes):**

1. The students will be able to analyze of series-series Feedback Amplifiers.

**Use of appropriate methods:**

**Justification for choosing the topic using One Minute Paper activity:**

Series-series Feedback Amplifiers is analysed based on the parameters such as gain without feedback, feedback factor, gain with feedback and input and output resistance which all involves many equations. As students may get confused, it is better to give practice on writing expressions. So that they can be able to recall all related equations.

**Effective presentation**

**Implementation (Plan & Execution) with Proof:**

- ✓ Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally.
- ✓ I asked all students to take a piece of paper and made them to write equations whatever they remember related to given topic.

## Sample Images:

**UNIT I**  
**Analysis of series-series Feedback Amplifiers**

95368106022  
ECE-A

One Minute  
activity

Current Series feedback

open loop gain =  $\frac{I_o}{V_s}$

$G_{m_{ol}} = \frac{I}{V}$

output current,  $I_o = -h_{fe} I_b$

IF voltage,  $V_o = I_b R_s + I_b h_{ie} + I_b R_E$

$G_m = \frac{-h_{fe} I_b}{I_b (R_s + h_{ie} + R_E)}$

$G_m = \frac{-h_{fe}}{R_s + h_{ie} + R_E}$

$\beta = \frac{V_o}{I_o} = \frac{R_E I_E}{I_o} = -\frac{R_E I_o}{I_o}$

$\beta = -R_E$  [ $\because I_E = I_o$ ]

A-Balo Kiruthika  
ECE-A

Electronic Circuit - II

$I_o = -h_{fe} I_b$

$V_o = R_s I_b + h_{ie} I_b + R_E I_b$

$G_m = \frac{I_o}{V_o} = \frac{-h_{fe}}{R_s + h_{ie} + R_E}$

$\beta = \frac{V_o}{I_o} = \frac{R_E I_E}{-I_E} = -R_E$

$\beta = -R_E$

Desensitivity,  $D = 1 + \beta G_m$

$D = 1 + (-R_E) \left( \frac{-h_{fe}}{R_s + h_{ie} + R_E} \right)$

$D = \frac{R_s + h_{ie} + R_E + R_E h_{fe}}{R_s + h_{ie} + R_E}$

$D = \frac{R_s + h_{ie} + R_E (1 + h_{fe})}{R_s + h_{ie} + R_E}$

tho  $G_{mf} = \frac{G_m}{D}$

### Significance of Results:

#### Assessment of Effectiveness/Success of the Activity:

- The students felt easy to remember and recollect the concepts and equations involved in the given topic which will make them to attend the questions under this topic in Internal Assessment Test.

#### Reflective Critique:

##### • Benefits:

- Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
- Making all the students to participate simultaneously
- Students can gather their thoughts prior to verbalizing them and benefits that student who is more fearful of public speaking by giving them a script to fall back on (or build on).

##### • Challenges:

- As it is mentioned above that it takes one minute to conduct this activity. But, the students are failed to remember all equations in a given duration. They took five minutes to complete this activity.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1            | 3   | 1   | 3   | 1   | 2   | -   | -   | 1   | -   | 2    | 1    | 1    |

CO1: The students will be able to analyze the various properties of signals and systems.

#### References:

- <https://provost.tufts.edu/celt/files/MinutePaper.pdf>
- Sedra and Smith, "Micro Electronic Circuits"; Sixth Edition, Oxford University Press, 2011.
- S.Salivahanan, N Suresh Kumar, "Electronic Circuits II" Mc Graw Hill Education.

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Electronics and Communication Engineering**  
**Academic Year: 2019- 2020 (Odd Semester)**

**Innovative Practices Description**

**UNIT III Linear Time Invariant – Continuous Time Systems**

**Degree, Semester & Branch:** III Semester B.E ECE A

**Course Code & Title:** EC8352 Signals and Systems

**Name of the Faculty member:** Mrs.R.Chitra

**Name of the Topic:** Impulse response, Convolution Integral

**Name of the Innovative Practice:** Quiz Dominos

**Date & Duration:** 19.08.19 & 15 minutes

**Description:**

Dominoes – key words/terms and their definitions are written on cards given to the group of students. A key word/term is read out and the one person with the correct definition for this key word/term from some other group stands up and reads out their definition and so on... this continues in a loop until you are back to the start – great fun when you put learners against the clock.

**Goals (Learning Outcomes):**

- The students will be able to apply the properties of impulse response and convolution integral to analyze Linear Time Invariant CT Systems.

**Use of appropriate methods:**

**Justification for choosing the topic using Quiz Dominos activity:**

A convolution is a mathematical way of combining two signals to form a third signal which is used to analyze the LTI (Linear and Time-Invariant) system or filter. Students must know about this topic. But, it is difficult to study for students. To make the students feel easy with this topic, I conducted this activity (Quiz Dominos) for this topic which made the students to understand better.

**Effective presentation:**

**Details of the Implementation:**

- I have prepared some keywords/questions related to convolution integral and its properties and wrote one key word/question per chart paper (small).
- Made all students to sit in a team(3-4 students)
- Distributed one chart paper per team
- Initially I asked one team to read given key word in the given chart paper
- The one student with the correct definition for this key word/term from some other team stands up and reads out their definition and so on... this continues in a loop until all questions are read and find out its answer

## Sample Images



## Reflective Critique:

### Benefits:

- Students were able to understand easily
- Made entire class to be fully engaged during execution of the activity
- Can be improved their presence of mind
- Had a great fun when i put learners against the clock.

### Challenges:

- Planned 15 minutes but it took 20 minutes to complete this activity

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO3            | 2   | 3   | 2   | 1   | 1   |     |     | 1   |     | 2    |      | 2    |

*CO3: The students will be able to analyze linear time invariant continuous time systems using Laplace and Fourier Transforms.*

### References:

1. <https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/video-lectures/lecture-4-convolution/>
2. <https://handsonaswegrow.com/dominoes/>

**RAMCO INSTITUTE OF TECHNOLOGY**  
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**Academic Year: 2019- 2020 (Odd Semester)**

**Innovative Practices Description**

**UNIT IV Analysis of Discrete Time Signals**

**Degree, Semester & Branch:** III Semester B.E ECE A

**Course Code & Title:** EC8352 Signals and Systems

**Name of the Faculty member:** Mrs.R.Chitra

**Name of the Topic:** Z transform

**Name of the Innovative Practice:** Think-Pair-Share

**Date & Duration:** 19.09.19 & 15 minutes

**Description:**

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to (1) think individually about a topic or answer to a question; and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

**Goals (Learning Outcomes):**

- The students will be able to analyze discrete time signals using Z transform.

**Use of appropriate methods:**

**Justification for choosing the topic using Think-Pair-Share activity:**

The Z Transform has a strong relationship to the DTFT, and is incredibly useful in transforming, analyzing, and manipulating discrete calculus equations in the field of digital signal processing. It will be useful to study Digital Signal Processing subject in V Semester to analyze digital filters and to analyze the linear discrete system. So, I have chosen this topic to conduct this activity for making students to share their ideas among their peers.

**Effective presentation**

**Details of the Implementation:**

- I have planned this activity for 15 minutes
- Initially I have explained what students need to try and asked the students to form a pair
- Made all the students to engage in participation
- Students were discussed about the given topic with each other in talk partners for five minutes.
- After that, they are asked to express their thoughts to an audience and contribute to a shared solution.
- Make sure there is enough time for both students in a pair to discuss their ideas.

### Sample Images:



### Reflective Critique:

#### Benefits:

- This step ensures that every student has the opportunity to explain their thoughts to others and listen to others conversations.
- Due to this activity, even though one student leaves any point the other student will help to recollect the ideas. So they felt comfortable to think back and share it.
- Students were able to remember the content of the topic by conducting this activity and maximum students were attended the questions under this topic in Internal assessment Test.

#### Challenges:

- One of the biggest challenges of the think-pair-share is to get all students to truly be engaged
- Most pairs shared the same ideas about the topic
- It took more time than what I planned

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO4            | 2   | 3   | 2   | 1   | 1   |     |     | 1   |     | 2    |      | 2    |

*CO4: The students will be able to analyze discrete time signals using DTFT and Z transform.*

#### References:

1. Poorna Chandra S, Sasikala. B.Digital Signal Processing, Vijay Nicole/TMH, 2013.
2. [www.et.iitb.ac.in](http://www.et.iitb.ac.in) > Resources > files > TPS



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**Innovative Practices Description**

**UNIT V Linear Time Invariant - Discrete Time Systems**

**Degree, Semester& Branch:** III Semester B.E.ECE A

**Course Code & Title:** EC8352 Signals and Systems

**Name of the Faculty member:** Mrs.R.Chitra

**Name of the Topic:** Convolution Sum

**Name of the Innovative Practice:** Sorting Strips

**Date & Duration:** 30.09.2019 & 20 minutes

**Description:**

Small bits of information are separated into strips so that students can sort the strips into various categories, or organize them into a sequence depending on the topic. This strategy encourages discussion of competing ideas or organizations or order in which a process would take place. In this case, it is often the discussion and sharing of ideas that is the most important outcome of the activity.

**Goals (Learning Outcomes):**

- The students will be able to analyze Linear Time Invariant Discrete time systems using convolution sum.

**Use of appropriate methods:**

**Justification for choosing the topic using Sorting Strips activity:**

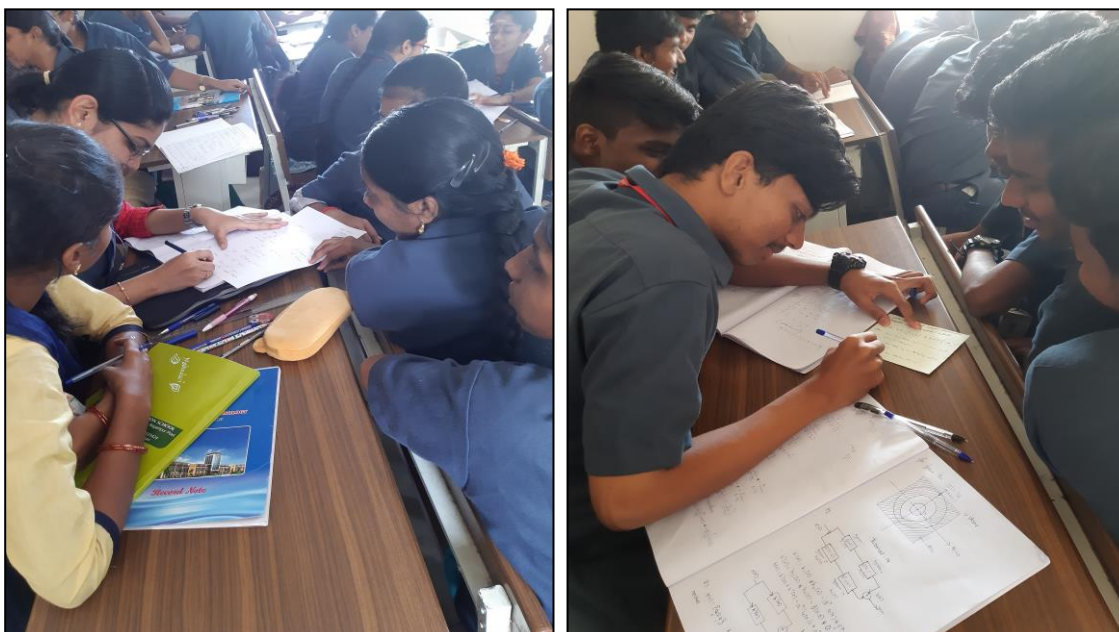
A convolution is a mathematical way of combining two signals to form a third signal which is used to analyze the LTI (Linear and Time-Invariant) system or filter using Z transform. Students must know about this topic. But, it is difficult to study for students. It will be useful to analyze digital filters and to analyze the linear discrete system. So, I have chosen this topic to conduct this activity for making students to share their ideas among their peers.

**Effective presentation:**

**Details of the Implementation:**

- The goal of this activity is for students to order a set of items, such as steps for solving problems.
- For this, I made entire class into smaller groups (4 to 5 students) and given pieces of papers to each group.
- As one strategy, I given students with a list of procedures for performing convolution operation written on strips of paper that are jumbled
- Students were asked to work together to reconstruct the proper sequence.

### Sample images:



### Significance of Results:

#### Assessment of Effectiveness/Success of the Activity:

Usually the students don't attend this topic convolution sum since it is a lengthy procedure. But after conducting this activity they got a clear idea about this topic and most of the students attended this question asked in Internal Assessment Test III.

#### Reflective Critique:

##### Benefits

- This activity helps students apply what they have learned through reading or didactic teaching.
- This approach can strengthen students' logical thinking processes and test their mental model of a process.
- In this case, it is often the discussion and sharing of ideas that is the most important outcome of the activity.

##### Challenges

- I planned to conduct this activity in 15 minutes. But it took around 20 minutes.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO5            | 2   | 3   | 2   | 1   | 1   |     |     | 1   |     | 2    |      | 2    |

CO5: The students will be able to interpret the linear time invariant discrete time systems using DTFT and Z transform.

#### References:

1. <https://blogs.baruch.cuny.edu/saccactivelearning/>
2. <https://teaching.berkeley.edu/active-learning-strategies>
3. <http://www.wlxt.uestc.edu.cn/wlxt/ncourse/dsp/web/kj/Chapter2/2.6.1%20Convolution%20Sum.htm>